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Text-as-Data/Natural Language Processing for Political Science (MA Political Science)

Wednesdays, 2:00–6:00 PM (c.t.), bi-weekly – (Start: 23.04.2025; End: 16.07.2025)
Room: 027, Oettingenstraße 67

Office hours by appointment via Zoom or at the Geschwister-Scholl-Institute of Political Science (Room U168, Oettingenstraße 67)

Course description

What messages do political actors convey on social media and in parliament? What topics dominate the media agenda? What is the content of laws? Today, political actors and institutions generate more texts than ever before. Analysing these texts is crucial for our understanding of democracies, especially in the age of digitalisation and big data. Computational methods like Text-as-Data/Natural Language Processing (NLP) techniques are essential for analysing vast amounts of political text, offering deeper insights into political behavior, policy trends, and public discourse.

The course provides an introduction to text-as-data/natural language processing for political science research. Participants will gain insights into the current state of the field and practical experience in designing and implementing quantitative text analysis methods. The course teaches students the necessary steps to conduct their own text-as-data/natural language processing research projects: (1) setting up a data collection and processing pipeline, (2) applying various text analysis methods (e.g., topic classification, scaling, sentiment analysis), and (3) presenting, discussing, evaluating, and interpreting the results.

Note: Students are required to bring their own laptop for the practical exercises.

Course requirements

- Submission of all exercises (not graded) is required to pass the course.
- Students must develop a term project (in pairs) that addresses a political science research question using Text-as-Data/Natural Language Processing methods. The project should demonstrate both an understanding of these methods and the ability to apply them for political science research.
- The successful completion of the term project (minimum grade: 4.0) is required to pass the course.

Required readings and course materials

The readings listed in the schedule are mandatory for all participants as preparation for each session. The required readings and any additional materials are available on the Moodle platform: https://moodle.lmu.de/course/view.php?id=38778

The course will, in particular, build on the following resources:

- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press.
- Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). *Computational Analysis of Communication*. Wiley Blackwell. https://cssbook.net/

Exercises

Students are required to complete take-home exercises after each session. These exercises will be provided as .ipynb files (e.g., *exercise_01.ipynb*) on Moodle, in the section corresponding to the respective session.

During the sessions, students will begin working on the exercises in lab sessions and are expected to complete the remaining tasks over the following week. The deadline for submission is always the Wednesday of the following week at 11:59 PM.

The exercises are designed to provide a practical introduction to the topics covered in class, helping students gain hands-on experience in implementing methods and techniques for political text analysis.

The exercises are not graded, but submission of all exercises is required to pass the course!

Term project

Students must develop a term project (in pairs) that addresses a political science research question using Text-as-Data/Natural Language Processing methods. The project should demonstrate both an understanding of these methods and the ability to apply them for political science research.

The term project should address a relevant research puzzle and research question, provide a (short) literature review and theoretical framework, and present a research design in which text analysis is the central methodological approach. This includes a detailed description of how text analysis methods are applied to answer the research question. The project should also present results and conclusions while critically reflecting on the methodological choices. A particular emphasis should be placed on discussing the advantages and limitations related to the selection and collection of political texts, the processing and representation of texts, the choice of text analysis methods and models, and the validity of the results.

To help with the progress of the term project, students are required to submit an abstract (max. 250 words) on 28 May 2025, and a proposal (max. 1,500 words) on 25 June 2025. The course instructor will provide feedback on both submissions. Additionally, students will have an opportunity to work on their projects during the session on July 2, 2025.

On July 16, 2025, students will present their projects and receive valuable feedback from both their peers and the instructor. Students are also encouraged to use office hours for additional guidance throughout the project.

The deadline for the submission of the **term project paper** (5,000-6,000 words) and accompanying **replication material** (.ipnby-file) is 31 August 2025 at 11:59 PM. The successful completion of the term project (minimum grade: 4.0) is required to pass the course!

Aims of the course

- Understand how political texts (e.g., social media posts, parliamentary speeches, news articles) can be used for political science research.
- Gain knowledge about Text-as-Data techniques and methods and learn about their respective advantages and limitations.
- Develop practical skills in collecting, processing, and analysing textual data for political science research.
- Gain experience in presenting, interpreting, and evaluating findings from political text analysis.

Overview of individual sessions and exercises

Session/ Exercise	Date/ Deadline	Topics and Tasks
SE01	23.04.2025	• Introduction: Course plan and course requirements • Theory and background: Motivation and introduction • Lab session: Technical setup and first steps (*exercise_01.ipynb*)
EX01	30.04.2025	• Submission of *exercise_01.ipynb* via Moodle
SE02	07.05.2025	• Theory and background: Data selection and data collection • Lab session: Text corpora, scraping online data, data wrangling (*exercise_02.ipynb*)
EX02	14.05.2025	• Submission of *exercise_02.ipynb* via Moodle
SE03	21.05.2025	• Theory and background: Text representation • Lab session: Pre-processing, text cleaning, text-as-data (*exercise_03.ipynb*)
EX03	28.05.2025	• Submission of *exercise_03.ipynb* via Moodle • Submission of abstract for term project (max. 250 words) via Moodle
SE04	04.06.2025	• Theory and background: Classification - Classifying Documents into Known and Unknown Categories • Lab session: Dictionaries, supervised classification, unsupervised classification (*exercise_04.ipynb*)
EX04	11.06.2025	• Submission of *exercise_04.ipynb* via Moodle
SE05	18.06.2025	• Theory and background: Scaling - Measuring latent features in political texts • Lab session: Supervised scaling, Unsupervised scaling (*exercise_05.ipynb*)
EX05	25.06.2025	• Submission of *exercise_05.ipynb* via Moodle • Submission of proposal for term project (max. 1,500 words) via Moodle
SE06	02.07.2025	• Current trends and developments • Lab sessions: Work on term projects
EX06	09.07.2025	• Submission of *feedback.docx* via Moodle
SE07	16.07.2025	• Presentation of term projects
—	**31.08.2025**	• **Submission of term project via Moodle**

Content of individual sessions and exercises

(SE01) 23 April 2025 – Introduction

What are Text-as-Data/Natural Language Processing methods? Why are they useful and relevant for political science?

Course plan and course requirements

- Course description
- Course requirements
- Required readings
- Course work
- Overview of individual sessions and exercises
- Aims of the course
- Moodle

Theory and background: Motivation and introduction

- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. (Chapters 1 and 2)

Lab session: Technical setup and first steps

- Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). Computational Analysis of Communication. Wiley Blackwell. https://cssbook.net/. (Chapters 1, 3, 4 and 5)
- Task: *exercise_01.ipynb*

(EX01) 30 April 2025 – Submission of *exercise_01.ipynb* via Moodle

(SE02) 07 May 2025 – Data selection, collection and management

How do we select, collect, and ensure the quality of text data for political science research? How can we manage (large-scale) political text data?

Theory and background: Data selection and data collection

- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. (Chapters 3 and 4)

Lab session: Text corpora, scraping online data, data wrangling

- Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). Computational Analysis of Communication. Wiley Blackwell. https://cssbook.net/. (Chapters 6, 7 and 12)
- Task: *exercise_02.ipynb*

(EX02) 14 May 2025 – Submission of *exercise_02.ipynb* via Moodle

(SE03) 21 May 2025 – Text representation and text processing

How can we represent text data for analysis? What key processing steps are needed to clean and prepare text data for analysis?

Theory and background: Text representation

- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press. (Chapter 5; optional: Chapter 8)

Lab session: Pre-processing, text cleaning, text-as-data

- Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). Computational Analysis of Communication. Wiley Blackwell. https://cssbook.net/. (Chapters 9 and 10)
- Task: *exercise_03.ipynb*

(EX03) 28 May 2025 – Submission of *exercise_03.ipynb* & abstract for term project (max. 250 words) via Moodle

Abstract for term project: Important elements

- Research topic and relevance
- Research puzzle and question
- Theories and concepts
- First ideas for research design:
    - Text selection and collection
    - Text analysis methods and tools
- Expected results and contributions

(SE04) 04 June 2025 – Classification

What insights can document classification provide in political analysis? What are the main methods for classifying political texts, and how do they work? How can we ensure accuracy and minimise bias in document classification?

Theory and background: Classification - Classifying Documents into Known and Unknown Categories

- Grimmer, J., & Stewart, B. M. (2013). Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts. *Political Analysis, 21*(3), 267–297. https://doi.org/10.1093/pan/mps028

Lab session: Dictionaries, supervised classification, unsupervised classification

- Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). Computational Analysis of Communication. Wiley Blackwell. https://cssbook.net/. (Chapter 11; optional: Chapter 8)
- Task: *exercise_04.ipynb*

(EX04) 11 June 2025 – Submission of *exercise_04.ipynb* via Moodle

(SE05) 18 June 2025 – Scaling

What insights can document scaling provide in political analysis? What are the main methods for scaling political texts, and how do they work? How can we ensure accuracy and minimise bias in scaling?

Theory and background: Classification - Measuring latent features in political texts

- Grimmer, J., & Stewart, B. M. (2013). Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts. *Political Analysis, 21*(3), 267–297. https://doi.org/10.1093/pan/mps028

Lab session: Supervised scaling, unsupervised scaling

- Task: *exercise_05.ipynb*

(EX05) 25 June 2025 – Submission of *exercise_05.ipynb* & proposal for term project (max. 1,500 words) *via* Moodle

Proposal for term project: Important elements

- Research topic and relevance
- Research puzzle and question
- Theories and concepts
- Hypothesis
- Description and reflection of research design:
    - Text selection and collection
    - Text analysis methods and tools
    - Validation
- Expected results and contributions
- Time plan

(SE06) 02 July 2025 – Current trends and developments

What are current trends and developments in the field of text-as-data and Natural Language Processing for political science? How are advances in machine learning and AI shaping text analysis in political research?

Theory and background: Replication and open data, LLMs (e.g., ChatGPT), multimedia data

- Alizadeh, M., Kubli, M., Samei, Z., Dehghani, S., Zahedivafa, M., Bermeo, J. D., Korobeynikova, M., & Gilardi, F. (2025). Open-source LLMs for Text Annotation: A Practical Guide for Model Setting and Fine-tuning. *Journal of Computational Social Science, 8*(1), 1-25. https://doi.org/10.1007/s42001-024-00345-9
- Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks. *Proceedings of the National Academy of Sciences, 120(*30): e2305016120. https://doi.org/10.1073/pnas.2305016120
- Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). Computational Analysis of Communication. Wiley Blackwell. https://cssbook.net/. (Chapters 14 and 15)

Lab session: Work on term projects

(EX06) 09 July 2025 – Submission of *course feedback* via Moodle

(SE07) 16 July 2025 – Presentation of term project

(—) 31 August 2025 – Submission of term project via Moodle

Term project – paper (5,000-6,000 words): Important elements

- Title
- Abstract (150-250 words)
- Introduction:
  - Research topic and relevance
  - Literature review
  - Research puzzle and question
  - Summary of research design
  - Summary of findings
- Theoretical framework
  - Theories and concepts
  - Hypothesis
- Research design:
  - Text selection and collection
  - Text analysis methods and tools
  - Validation
  - Discussion of advantages and limitations
- Results
  - Presentation of findings (figures, tables, etc.)
  - Interpretation and explanation of results
- Discussion and conclusion
  - Summary of central findings
  - Interpretation of findings in the broader research context
  - Limitations
  - Avenues for future research
- References
- Appendices (if needed)

Term project – replication material (.ipynb-file)

Grading scheme for term project

Grading category	Points
Argumentation and literature: Does the submission present a clear, logically structured argument, supported by a wide range of relevant and high-quality sources? Does it demonstrate critical engagement with the literature, highlighting key debates and perspectives in the field?	… / 20
Research design: Is the research design clearly explained and well-reflected upon? Does it demonstrate a sound methodology, with appropriate case selection, methods and a clear rationale for their use? Does the design align with the research question and objectives?	… / 20
Text-as-Data/NLP: Does the submission apply text-as-data or NLP techniques appropriately? Are these techniques clearly explained and reflected upon? Is the implementation of these techniques well-executed, and are their strengths and limitations considered in the analysis? Does the use of these techniques contribute meaningfully to answering the research question?	… / 20
Structure and language: Is the text clearly organised, with a logical flow (e.g., introduction, theoretical framework, research design, results, discussion and conclusion)? Does the submission follow academic conventions in terms of language, tone, and formatting? Is the writing precise, formal, and free from grammatical errors, making the argument accessible to an academic audience?	… / 10
Citations and referencing: Are sources cited correctly throughout the paper, following an appropriate citation style? Is the bibliography complete, properly formatted, and consistent with the citation style? Does the paper avoid plagiarism by appropriately acknowledging all sources of information and ideas?	… / 10
Replication material: Is the submission accompanied by complete, functioning, and well-documented replication material (e.g., code, data, instructions)? Does the material allow others to reproduce the research findings reliably? Is the documentation clear, thorough, and user-friendly, making the replication process accessible to others?	… / 20
Overall points	… / 100